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Author(s): Charles R. Plott and Vernon L. Smith

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An Experimental Examination of Two Exchange Institutions

CHARLES R. PLOTT
California Institute of Technology

and

VERNON L. SMITH
University of Arizona

1. INTRODUCTION

Economists have shown increasing interest in the analysis of information and organization in decentralized price adjustment processes. This interest is most frequently manifested at the theoretical level (summaries of the relevant literature, concepts, and methodologies have been provided by Hurwicz [3] and Rothschild [4]), but it also has found expression through several laboratory experimental studies. These latter have unusual appeal because they can serve as a “proving ground” for the former. If well-formulated theories consistently fail to predict simple laboratory behaviour, then one would be hesitant to trust their predictions in richer environments. Furthermore, when replicable laboratory behaviour can be demonstrated, one should seek those extensions of accepted theory which explain why it occurs.

The laboratory studies have attempted to measure the effect of different forms of market organization on the efficiency, performance, convergence pattern and equilibrium level of prices. The variables of market organization examined have been the mechanisms or rules under which prices are communicated and contracts are executed. The focus on these particular variables has been motivated by the more prominent features of certain observed exchange institutions rather than formal models based on pre-scientific, casual observations of the kind that form the basis of embryonic modelling. We view the present work as providing a rigorous empirical foundation for a more formal development of theory.

2. TRADING INSTITUTIONS

The experiments we report here are based on two modes of trading organization.

(a) The one-sided oral auction in which buyers (sellers) make repeated oral price bids (offers) for the exchange of many units, one unit at a time. This form of market organization has been studied by Smith [6] for the case in which each of many traders has a capacity to exchange a single unit of the commodity per trading period. This process has the important informational characteristic that traders obtain sequential price observations as binding contracts are formed. The “country auction” for livestock and machinery is typically organized in this manner.

(b) The posted-price institution in which the price quotations cannot be altered (at least not without considerable cost) during the exchange period. Williams [9] studied the case in which buyers (sellers) of multiple units each post one price bid (offer) and sellers (buyers) respond with quantity offers which are then accepted in whole or in part by the buyers (sellers). Williams' [9] objective was to adapt the one-sided studies of Smith [6] specifically to ordinary retail markets and other price-posting markets, in which there is a price commitment extending over several transactions. This modification of the experi-

mental design, we argue, is of considerable substance and is not simply a minor variation on trading mechanics.

In order to help understand the divergence of experimental results under these two forms of market organization, we will need to refer to the sealed-bid auction. In this institution, studied by Smith [7] and Belovicz [1], in each market period a single seller offers a quantity, Q , and invites buyers to submit independent bids at stated prices for stated quantities of the good. The bids are arranged from highest to lowest and the first Q bids are accepted. There are two variations:

- (i) In the *discriminative auction* all the accepted bids are filled at the stated bid prices. The best-known continuing market with this structure is the weekly auction of US Treasury bills.
- (ii) In the *competitive auction* all the accepted bids are filled at the price represented by the lowest of the accepted bids. The French auction of new stock issues corresponds approximately to this paradigm.

3. DIVERGENT EXPERIMENTAL RESULTS

The principal result in Smith's [6] experiments with oral auctions incorporating one-unit incentives is for contract prices beyond the first trading period to be executed to the disadvantage of the side making the quotations. Thus, when only sellers are permitted to make offers, contract prices tend to be significantly below the contract prices resulting when both buyers and sellers make quotations, which in turn are below contract prices occurring when only buyers can make bids.

Williams [9] modified Smith's experiment in two respects.

1. Smith's subjects could each trade only one unit, i.e. buyers had a total revenue "schedule" consisting of a single point, and sellers a total cost "schedule" consisting of one point. Williams' subjects were given total revenue and total cost schedules for multiple units. Thus Williams induces a demand or supply schedule for each subject, rather than a demand or supply point, for each subject.

2. Price quotations were made orally in the Smith experiment and could be revised during a trading period if no acceptance occurred. Thus the market was an auction where each buyer (seller) orally quoted a bid (offer) to sellers (buyers) *en masse* and could change his bid (offer) if no one accepted. Williams, in contrast required each subject buyer (seller) to submit a single price bid (offer). Each price was posted and a seller (buyer), selected at random, then chose a buyer (seller) and offered to sell (buy) a stated quantity. The buyer (seller) then accepted all or any part (but at least one unit) of the quantity offered. If the buyer (seller) accepted the entire quantity, the next seller (buyer), again chosen at random, could attempt to sell (buy) to (from) that same buyer (seller) before choosing another buyer (seller).

Williams reports results which are the exact opposite of those reported by Smith [6]. That is, when buyers post bids Williams finds that contracts, beyond an initial learning period, tend to be *below* the theoretical supply and demand equilibrium and when sellers post offers, he finds that contracts tend to *exceed* this theoretical price. The cumulative distribution of trades for two of Williams' experiments are shown on Figure 1. These distributions, which indicate for each price p^* the total volume of trades which occurred at prices $p \geq p^*$, are typical of his results in general.

We regard the second modification as the crucial treatment variable that explains the results reported by Williams. Our suspicions about the importance of this second treatment variable arise from the institutional similarities between the discriminative sealed-bid and posted-bid processes and also from the results which have emerged from sealed-bid experiments.

In both the posted-bid and sealed-bid institutions a price bid must be made before the

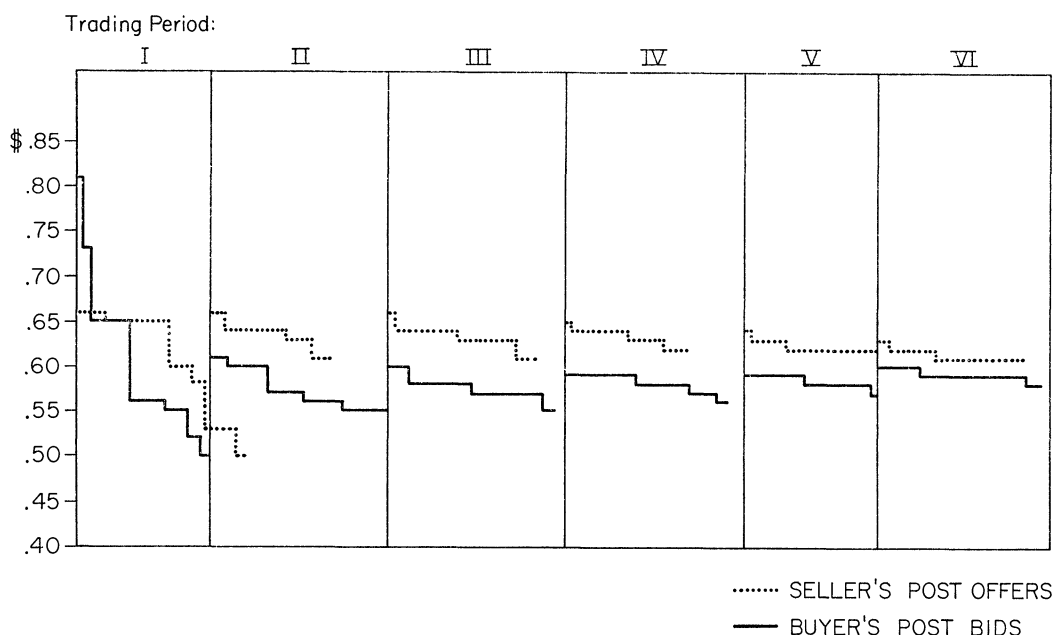


FIGURE 1

Williams' results: cumulative distribution of trades with multi-unit incentives. The figure represents for each price p^* , the volume of trades which took place at a price $p \geq p^*$

beginning of a period and without the knowledge of the bids made by other people. In the sealed-bid discriminative auction buyers know only the highest and lowest bids from previous periods and the total quantity to be sold, whereas in the posted-bid case all previous bids are known. Thus, if this informational difference proves to be minor, the information on market conditions is approximately the same under both institutions at the time of decision and the mode of price expression is identical.¹

Smith's [7] study of the sealed-bid auction shows that expected utility-maximizing buyers will tend to bid less in a discriminative auction than in a competitive auction. In the competitive sealed-bid auction, all bids are filled at the lowest of the accepted bids, while in the discriminative auction you pay what you bid. Hence, in the competitive auction there is no profit penalty in bidding above the low accepted bid, and so bidding one's true maximum willingness to pay (or limit price) maximizes the chance of being among the accepted bids. In the discriminative auction the ideal bid is epsilon above the bid one believes will be the lowest one to be accepted, and is therefore likely to be below one's true maximum willingness to pay. These theoretical predictions are supported by the experimental results of Smith [7] and are reinforced and considerably extended in the definitive study of sealed-bid auctions by Belovicz [1]. The same theory applies to the sealed-offer auction and leads to the prediction that sellers will tend to submit higher offer prices in a discriminative auction than in a competitive auction. Consequently, one should expect prices in a sealed-offer discriminative auction to dominate those in the sealed-bid discriminative auction. From this conjecture and the close similarity between the posted-price and discriminative auctions the Williams [9] experimental results are to be expected.

This argument suggests that there is no contradiction between Smith's [6] oral auction results and Williams' [9] posted-price results because Williams' examination of the multi-unit case was executed within institutions which are substantially different from those examined by Smith [6]. On this interpretation Williams' results take on an added dimension of significance.

A natural way to test our interpretation of the divergent results of Smith and Williams is to conduct new experiments comparing the multiunit oral-bid auction with the multiunit posted-bid organization. *A priori* we expected the distribution of prices in the oral-bid auction to dominate the distribution of posted-bid contracts. This was because in [7] Smith showed that expected utility-maximizing bidders would bid their true marginal valuations (limit prices) in a competitive sealed-bid auction. This implies that the competitive sealed-bid auction should yield results very close to that of the oral double auction. But from [6] it is known that the distribution of prices in the oral-bid auction dominates the distribution of prices in the double auction. Hence, we expect prices in the oral-bid auction to dominate those in the discriminative sealed-bid auction, and *pari passu* those in the similar posted-bid institution.

4. EXPERIMENTAL DESIGN

A total of four experimental sessions were conducted. Each of the four sessions involved four buyers and four sellers. Figure 2 shows the individual (marginal cost) induced supply, and (marginal revenue) induced demand functions implied by the total cost and total revenue schedules given to each subject. In each experiment two buyers (sellers) received schedules corresponding to $D_1(S_1)$, and two buyers (sellers) received schedules corresponding to $D_2(S_2)$. No subject participated in both experiments. All subjects were undergraduate students at Caltech who volunteered to participate in a "decision-making experiment" in which they would be paid US currency in amounts that would depend upon their decisions.

Experiments 1 and 3 are posted-bid procedures for buyers. The purpose of these experiments is to check the extension of the Williams results to our subject pool, experimental procedures, etc. Without such replication we cannot be sure that our research results are attributable to the treatment variable. Experiments 2 and 4 apply the oral-bid institution procedure used by Smith to the multiunit supply and demand conditions of experiments 1 and 3.

Experiments 1 and 2 differ from 3 and 4 in four important ways. First, in experiments 1 and 2 no commissions were paid for trading. This conformed with the incentive structure used by Williams. In experiments 3 and 4 a five-cent commission was paid for each trade. As will be explained below, this allowed greater accuracy in the efficiency comparisons between the two trading institutions. The remaining three differences are in the instructions, reproduced in Appendix 1. These differences reflect some evolution of experimental design in general since the Williams' experiment. First the format of the incentives differs. In experiments 1 and 2 buyers and sellers were given tables which contained the total and marginal costs and redemption values. Detailed forms for recording trades and profits were not used. Second, the time limit mentioned in experiment 4 was not used in experiment 2. Actually the time limit was not binding after the first trading period in experiment 4. The final difference is in the way bids were offered. In experiment 2 the experimenter called (in a rotating fashion) upon bidders who simultaneously signalled (by raising a hand) they wanted to bid. In experiment 4 bidders did not need to be recognized. They simply called out the bid—and the bid of the highest bidder was taken when ties in calling out occurred.

No economic sophistication on the part of the subjects was assumed. Indeed, two experimental sessions (experiment 0 and a first attempt at experiment 4) were discarded because of the failure of subjects, a seller in one case and a buyer in the other, to understand the incentive structure. In experiment 0 a seller sold triple the number of units that would have been profitable in the first trading period. This so depressed the market that several trading sessions were required for it to recover. This invalidated session was useful as a pilot test of our procedures and resulted in a decision to reduce the number of buyers and sellers by one-half and make the institutional and procedural changes indicated above. From extensive experience with these kinds of experiments we have learned that very small numbers with approximately equal power yield competitive market outcomes. A buyer

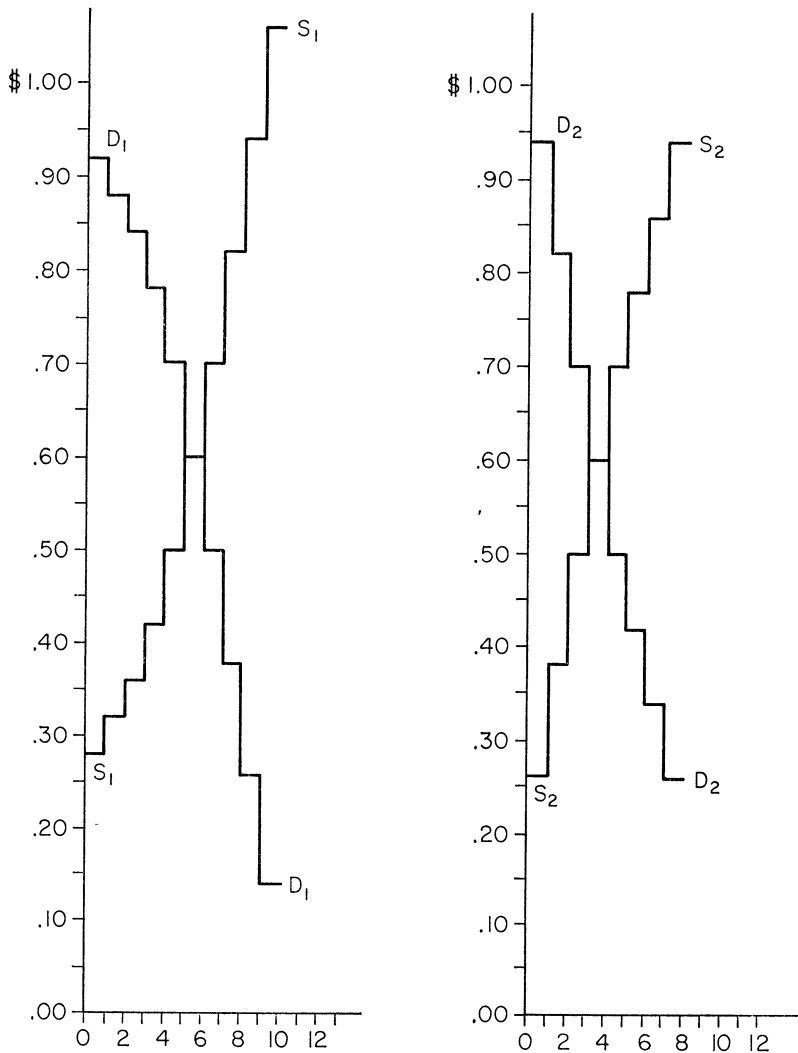


FIGURE 2
Induced individual supply and demand functions

in our first attempt to run experiment 4 purchased many more units than would have been profitable during the first period. The experiment was immediately terminated and later conducted with a new group of subjects.

5. EXPERIMENTAL RESULTS: PRICE COMPARISONS²

Our hypothesis is that the mode of organization chosen by Williams is a major treatment variable. This claim is supported by Figures 3 and 4 which report the cumulative distribution of contract prices for all four experiments.

Although we induced somewhat steeper supply and demand schedules than Williams [9, p. 10], the price distributions for our experiments 1 and 3 are very similar to those he reports for his buyer bidding sessions 2, 4 and 6. Through the first six to ten trading periods, the bids in all four of Williams' experiments appear to converge toward the

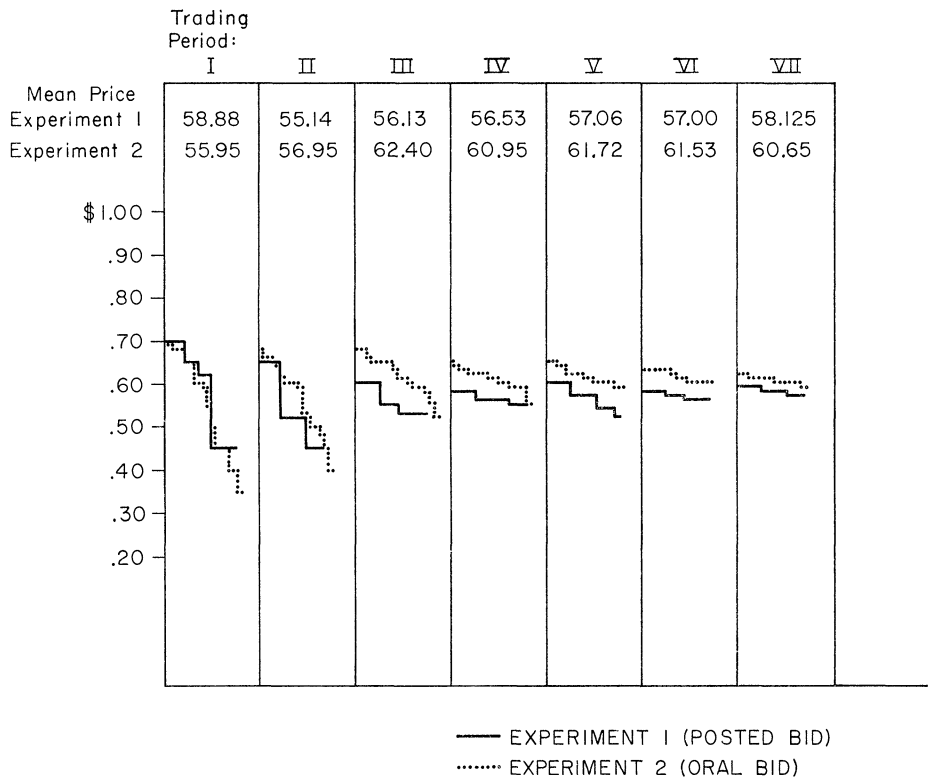


FIGURE 3

Cumulative distribution of trades. For the posted-bid institution this cumulative distribution is also the sequence of trades

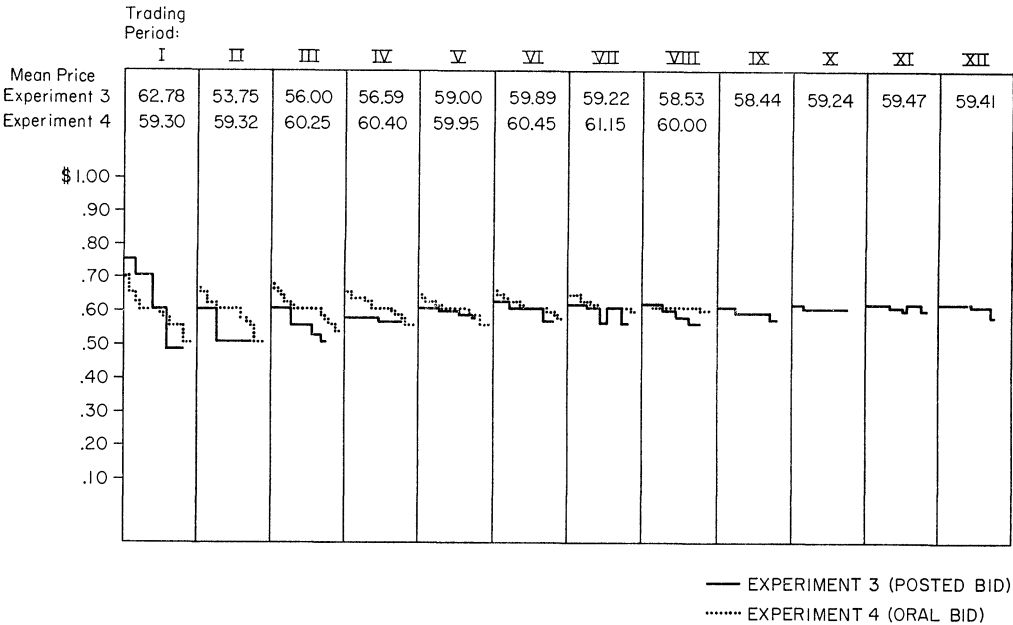


FIGURE 4

Cumulative distribution of trades. For the posted-bid institution this cumulative distribution of trades is also the sequence of trades in every period except XI. For period XI we report the sequence of trades

equilibrium price from below. However, Williams' session 6, which was conducted for twenty trading periods, reached equilibrium in the tenth period, but was below the equilibrium in the thirteenth to twentieth period. This suggests that some downward bias in bid prices may persist indefinitely.

It seems undeniable that we have replicated the research results of Williams' experiments with our subjects. The average price is below equilibrium in every period for both replications except period six in experiment 3, and prices appear to be converging to equilibrium from below. The quantity exchanged also tends to be below equilibrium.

In the oral bid experiments (2 and 4) for every period except the first, prices tend to be above the prices in the posted-bid institution. In addition there is a tendency for contracts to rise above the equilibrium price in the oral-bid auction. Hence, the Smith result generalizes, within this institution, to the multiunit case.

Williams' results appear to be a natural feature of "take-it-or-leave-it" pricing within which the sealed-bid and posted-bid institutions are special cases. The behaviour of sellers in the posted-bid institution is almost identical to the mechanistic response in the sealed-bid arrangement. Sellers respond to posted bids as pure competitors. They do not try to withhold supply, give preferential treatment or otherwise collude among themselves. They are evidently faced with a particularly difficult prisoner's dilemma. Of the eighty-one supply responses in the two experiments made by individual suppliers, only twenty were not the exact competitive response, and of these twenty only four can be interpreted as restrictions of supply (most of the remaining twenty offers exceed the competitive response with such subjects revealing a willingness to trade for less than the five-cent commission).

6. EXPERIMENTAL RESULTS: EFFICIENCY

Both the posted-bid and the oral-bid institutions are remarkably efficient (recall that no participant knows anything other than his own incentives), but the oral-bid auction is more efficient than is the posted-bid institution. One might suspect this from viewing Figures 3 and 4: volume in the oral-bid case is always greater than in the posted-bid case. This suggests that not all gains from trade are exhausted in the posted-bid institution.

These markets are perfectly efficient if and only if the maximum amount of money is extracted by the participants from the experimenter. If trades resulted in less than the maximum profit and commissions for the group as a whole, gains from trade would still be possible. The maximum total amount participants could receive in a period in experiments 1 and 2 is \$7.12, and in experiments 3 and 4 it is \$9.12, the difference, \$2.00, being due to commissions paid in experiments 3 and 4.

Where commissions are paid, it is easy to show that a market is efficient in a given period if and only if the following are true:

- (i) All buyers (sellers) exchange all units with redemption values (marginal cost) above (below) the competitive equilibrium price. These are called the *intra marginal demand (supply) units*.
- (ii) All buyers (sellers) exchange all units with redemption values (marginal cost) equal to the competitive equilibrium price. These are called *marginal demand (supply) units*.
- (iii) No buyers (sellers) exchange any units with redemption values (marginal cost) below (above) the competitive equilibrium price. These are called *extra-marginal demand (supply) units*.

The reader can get a good idea of the meaning of inefficiency and why inefficient trades occur in the posted-price institution by studying Figure 5. This is a sketch of the trading which occurred in period five of experiment 3. Buyers 17, 18, 19, 20, respectively, posted prices of 0.60, 0.59, 0.58, 0.57. Suppliers were ordered (by reference to a random number table) with 21 first, then 24, 23 and 22.

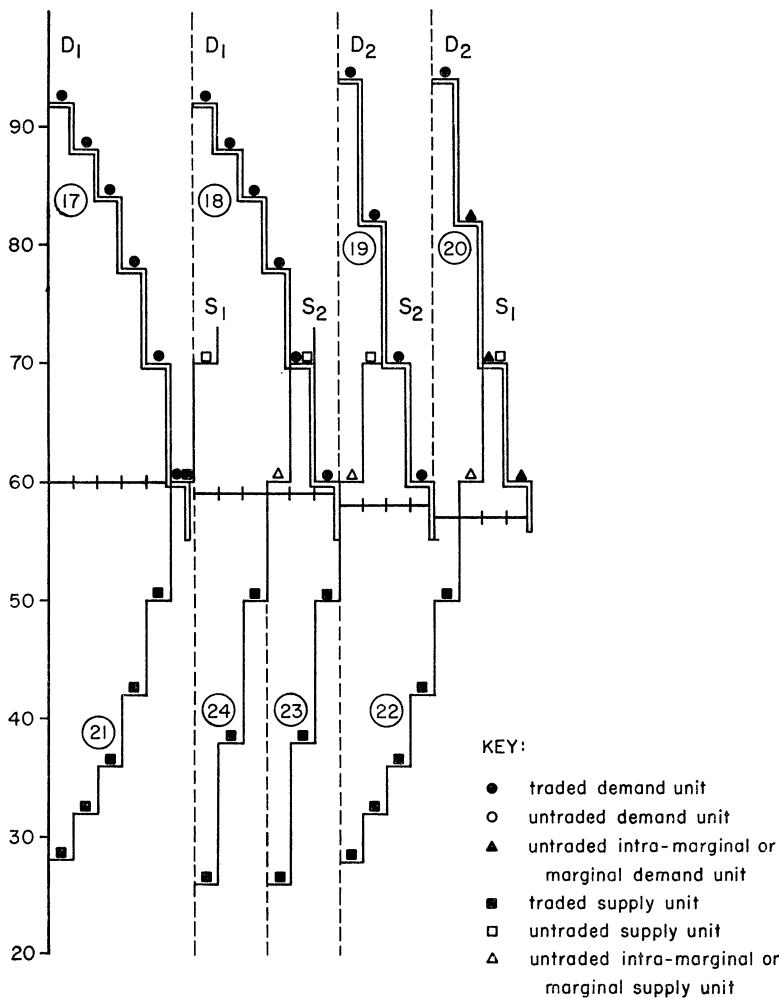


FIGURE 5
Contracts, Experiment 3, period V

Trading proceeded as follows. Seller 21 offered his competitive response of six units to the highest priced buyer, number 17, who purchased all six, his demand, and retired from the market. Next, seller 24 offered his competitive response of three units to buyer 18 who took them all. Then seller 23 offered his competitive response of three units to buyer 18 who purchased them all, then retired from the market. Seller 22, last in line, offered his competitive response of five units to buyer 19 who had bid 0.58. Buyer 19 purchased four of the five units offered, his quantity demanded, and then retired from the market. Seller 22, having sold five of his six units, then offered one unit to buyer 20 who took it.

The inefficiencies that resulted are as follows: Sellers 24, 23 and 22 failed to trade marginal units. Buyer 20 failed to trade two intramarginal units and one marginal unit. Profits amounted to \$6.80 for the period. Commissions came to \$1.70. Total return to participants was thus \$8.50, which is 93 per cent of the maximum.

Tables I and II summarize the effectiveness of these institutions to exhaust the gains from trade. Efficiency of the posted-bid institution averages 95 per cent across periods in experiments 1 and 3. Not bad, we would say, after considering the magnitude of the

TABLE I

Exchange efficiency of posted-bid institutions by period: Experiments 1 and 3

Trading period	Type of inefficiency by subject number			Efficiency measure
	(i) Untraded intramarginal units	(ii) Untraded marginal units	(iii) Traded extra-marginal units	
Experiment 1				
I	8	2, 3, 4, 5, 6, 8	7	0.97
II	4, 4, 4, 5, 7	2, 3, 4, 5, 6, 7, 8	—	0.92
III	4, 4, 4, 7	4, 5, 6, 7	—	0.94
IV	2, 2	2, 5, 6, 7	—	0.96
V	3, 3	3, 5, 7, 8	—	0.96
VI	2, 2	2, 3, 5, 6, 7, 8	—	0.99
VII	3, 3, 3	3, 5, 6, 7, 8	—	0.95
VIII	4, 4	4, 5, 6, 8	—	0.96
Experiment 3				
I	18, 22, 23	17, 18, 22, 23	20, 21, 24	0.91
II	20, 20, 20, 22	17, 20, 21, 22, 24	19	0.83
III	19, 19	19, 20, 21, 22, 23, 24	—	0.92
IV	17, 17	17, 22, 23, 24	—	0.94
V	20, 20	20, 22, 23, 24	—	0.93
VI	—	17, 20, 22, 24	—	0.98
VII	18	18, 21, 22	—	0.97
VIII	18, 18	18, 21, 23, 24	—	0.94
IX	19	19, 22, 24	—	0.97
X	17	17, 19, 22, 23, 24	—	0.96
XI	—	20, 22	—	0.99
XII	20, 20	20, 22, 23, 24	—	0.93

TABLE II

Exchange efficiency of oral-bid institution by period: Experiments 2 and 4

Trading period	Type of inefficiency by subject number			Efficiency measure
	(i) Untraded intramarginal units	(ii) Untraded marginal units	(iii) Traded extra-marginal units	
Experiment 2				
I	—	11, 12, 13, 14	—	1.00
II	9	9, 11, 13, 14, 15	—	0.99
III	—	9, 11, 13, 16	—	1.00
IV	—	10, 16	—	1.00
V	—	9, 11, 13, 16	—	1.00
VI	—	9, 10, 11, 13, 15, 16	—	1.00
VII	—	9, 10, 11, 13, 15, 16	—	1.00
Experiment 4				
I	—	26	25	0.99
II	—	26, 32	—	0.99
III	—	—	—	1.00
IV	—	—	—	1.00
V	—	—	—	1.00
VI	—	—	—	1.00
VII	—	—	—	1.00
VIII	—	—	—	1.00

problem. But relative to the oral bid, the posted bid is strictly second class. In both experiments 2 and 4 the oral-bid auction is over 99 per cent efficient in the first two periods and always 100 per cent efficient in the subsequent periods. An important implication is that the price bias in one-sided oral auctions affects income distribution but not resource allocation.

Of considerable interest, especially from a methodological point of view, is the importance of commissions in bringing trading in “tight” around the equilibrium. This can be seen by comparing the untraded marginal units in experiments 3 and 4 and reflects the possible importance of transactions costs in market environments (Smith [8]).

7. EXPERIMENTAL RESULTS: PRICE DYNAMICS

We have not yet formulated a satisfactory model of our experimental results. One cannot yet simply appeal to those laws of economics which account for the behaviour exhibited by these two institutions, but we will provide a detailed description of the contracting process.

The pattern of contracts is easy to summarize: in the oral-bid auctions prices rise to levels that are to the disadvantage of the side with price initiative and fall to levels that are advantageous to this side in the posted-bid case. Why should this be? If we look at the first-period contracts on Figures 3 and 4 we see there is a great deal of similarity between the behaviour of the two institutions. Since this similarity is not preserved in later periods, it is clear that a fundamental difference in the two institutions is in the dynamics of adjustment.

1. *Oral Bid*

Figures 6 and 7 present the pricing and bidding dynamics of oral-bid experiments 2 and 4. These patterns, when compared with the data reported by Smith [6, pp. 192-193], demonstrate the generality of the dynamics seen in experiments of this type. In each period prices tend to start below the equilibrium, rise above, then decline to equilibrium near the end of the period. In period 1 buyers, having no information as to prices that might be acceptable to sellers, start probing with bids below equilibrium. Sellers, not knowing that later bids may be better or whether they will succeed in contracting at later bids, accept some of these initial low bids. (Since there is excess demand at these bids not all of them *can* be accepted.) This may induce buyers to try still lower bids, which are not accepted (observe bids 3-5 in Figure 6; bid 8 in Figure 7). When bids stand unaccepted, buyers are likely to raise their bids to induce sales. Once this occurs, sellers are alerted to the effect that their “waiting” has in raising prices. Thereafter contract prices rise and go above equilibrium with many bids standing unaccepted (thirteen in Figure 6 before the peak at 70 and seven in figure 7 before the peak, also at 70). This price rise is arrested because with bids much above equilibrium there is excess supply and “waiting” sellers tend to break their silence with two- and three-way ties to accept bids. Buyers, sensing seller eagerness as revealed by tied acceptances, now lower their bids, and the final few contracts are closer to the equilibrium. This pattern tends to repeat itself in each period except that (1) the first contracts occur at higher prices in successive periods (the time sequence of opening contract prices is 40, 40, 52, 55, 59, 59, 62 in Figure 6; and is 50, 50, 55, 55, 55, 57, 59, 59 in Figure 7), and (2) the price rise peaks out at lower prices in successive periods (70, 68, 68, 65, 65, 63, 62 in Figure 6; 70, 66, 65, 65, 64, 65, 64, 61 in Figure 7).

We are fortunate in experiment 4 to have the opportunity to observe the process when very few unaccepted bids occur. From period three on, notice that rising price sequences occur without unaccepted bids. This was a particularly active oral auction, and buyers may have been competitively raising bids to avoid the non-acceptance experienced in the first two periods or simply to avoid bid congestion. Also it may mean that some vehicle in addition to unaccepted bids is operating to express the excess demand. The tendency to overshoot

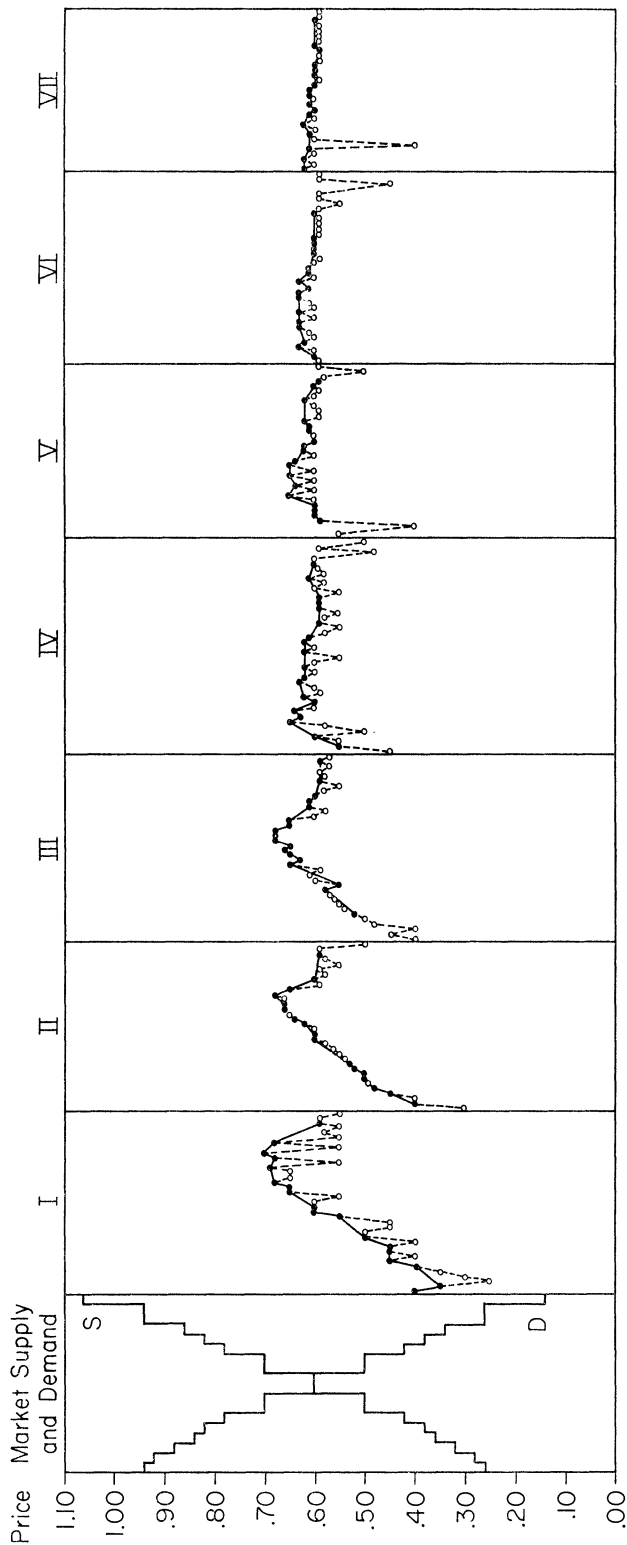


FIGURE 6
Sequences of bids and contracts. Experiment 2—(oral bid). ○ = unaccepted bid; ● = accepted bid (contract)

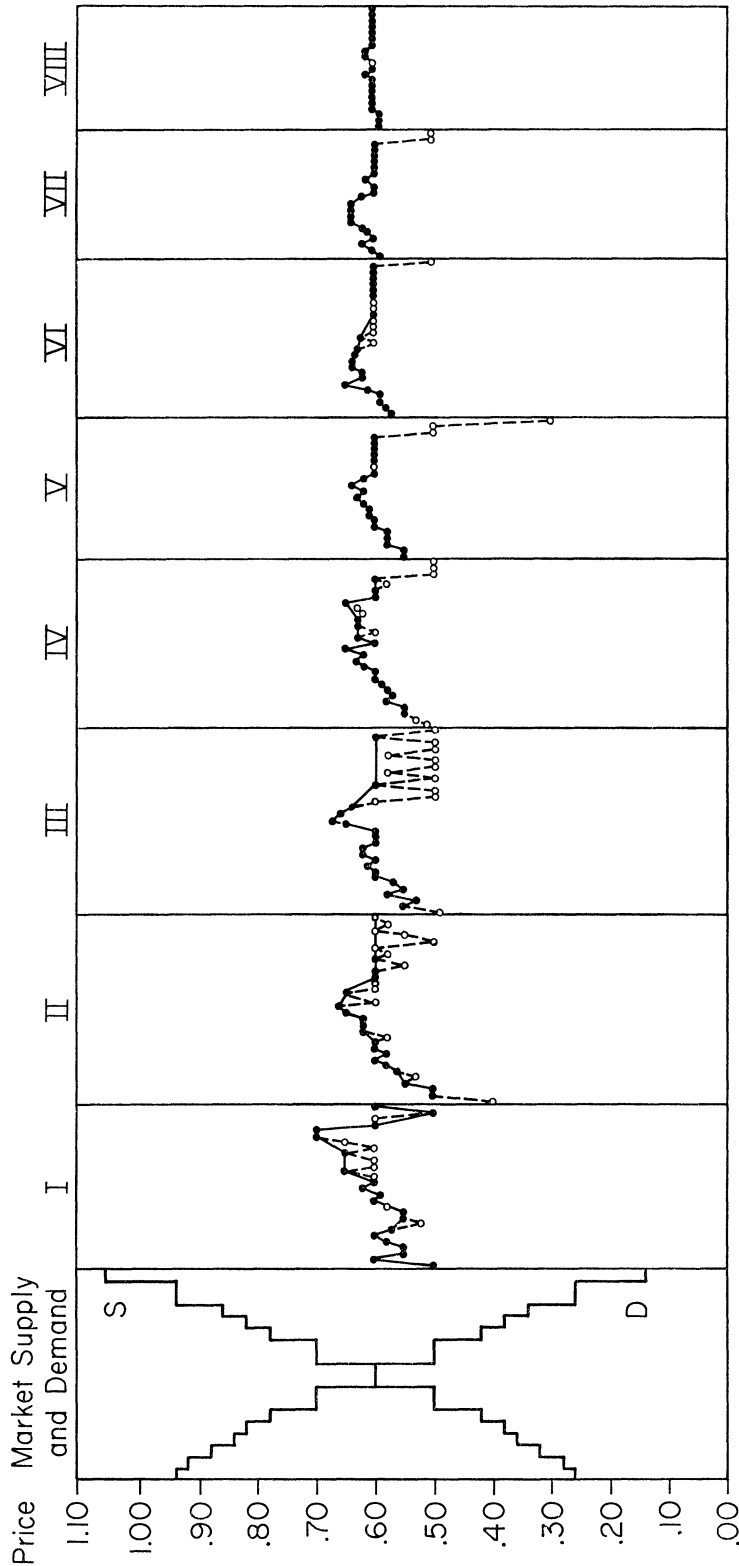


FIGURE 7
Sequences of bids and contracts. Experiment 4—(oral bid). ○ = unaccepted bid; ● = accepted bid (contract)

is lower in the absence of the unaccepted bids, thereby suggesting the importance of unaccepted bids as a partial explanation of the overshooting phenomenon.

2. Posted Bid

Figures 8 and 9 graph the sequence of bids entered by individual buyers in the two posted-bid experiments. In neither experiment is there a tendency for buyers to use changes in profits *alone* as a cue to future pricing decisions, so any model of the dynamics based on such a behavioural postulate would be suspect at best. For example, in experiment 1,

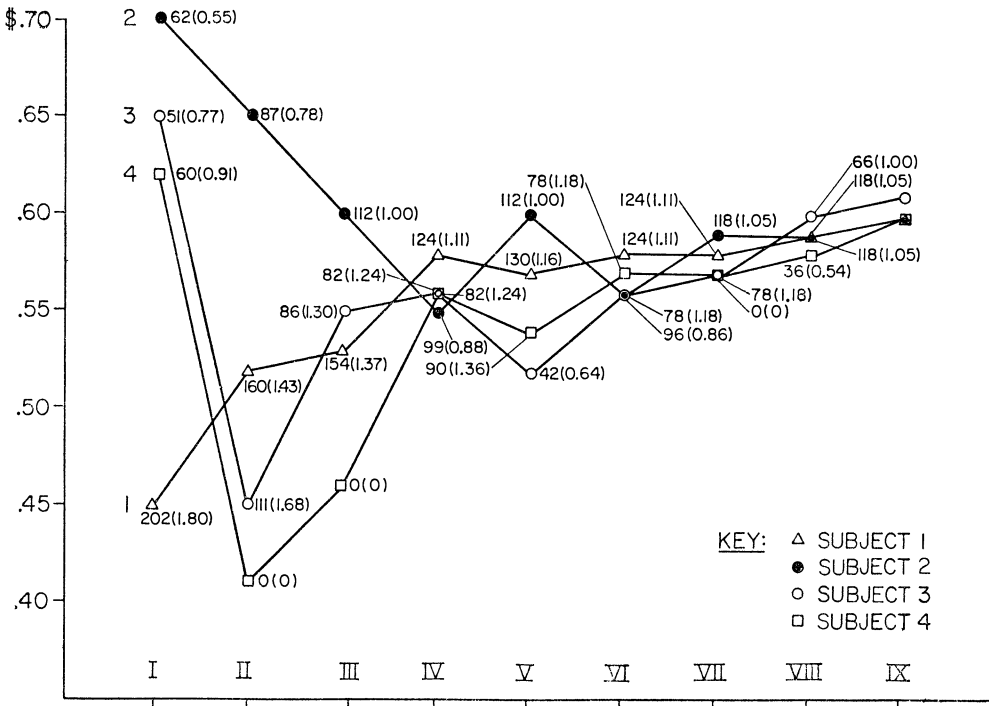


FIGURE 8

Buyer bids, profits, percentage of equilibrium profit by trading period. Experiment 1.

subject 1 increases prices for the first three periods even though it *seems* to produce reduced profits. He then lowers his price and experiences an increase in profits and raises prices for the rest of the experiment even though his profits continue to fall. Bidders experience two types of posterior errors in the announced price. It may be too high *or* too low. If it is the highest bid, then profits are lost if the bid is more than an epsilon above the next highest bidder. For example, in Figure 8 the high bidder always receives a smaller proportion of his potential profit than at least one other bidder. Subject 2 is severely penalized the first three periods by bidding too high. If a bid is the lowest, then, depending upon the distribution of remaining bids, sellers may satisfy all or most of their sales at the higher bids so that the volume for the low bidder is too small to yield a good profit (cf. Figure 9, period V). In experiment 1 the low bidder makes no purchases in periods II, III and VII, and only one in periods V and VIII.

The tendency for high and low bids to yield unsatisfactory profits has the expected effect on bid sequence as bidders begin to consider the actions of each other as well as their profits in formulating a bid strategy. In almost all periods of both experiments, the high

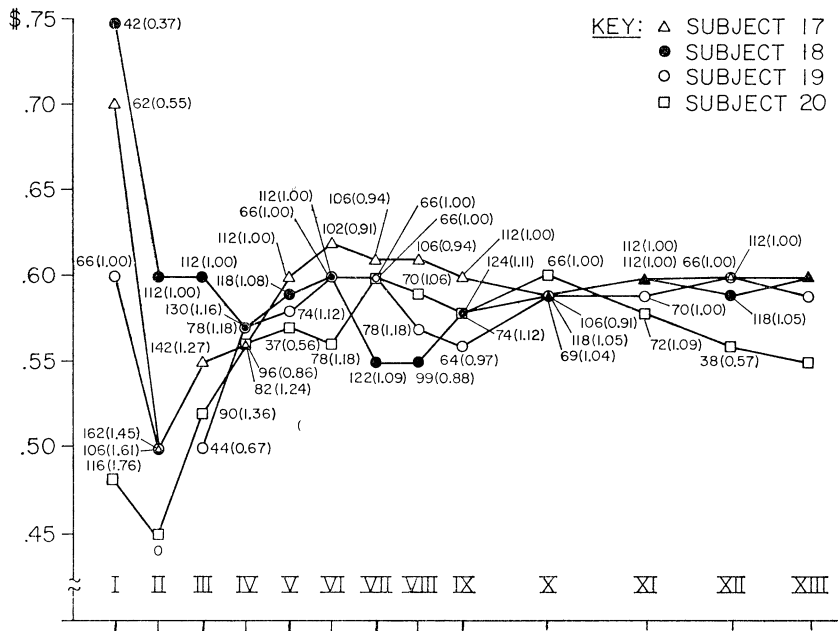


FIGURE 9
Buyer bids, profits, percentage of equilibrium profit by trading period. Experiment 3.

(low) bidder in period t lowers (raises) his bid in period $t + 1$. An exception occurs in periods XI, XII and XIII during experiment 3. Subject number 20 continues to lower his bid even in the face of greatly reduced profits. This, it seems, is a clear attempt at price leadership or tacit collusion. Most of our experience with these markets suggests that he will not have much success even though we suspect that the posted-bid institution provides a more favourable environment for co-operation than does the oral-bid auction.

8. SUMMARY AND CONCLUSIONS

We have studied two exchange institutions under experimental controls designed so that the experimental sessions were alike in all respects except that (1) experiments 1 and 3 (the “control” sessions) were conducted under irrevocable posted-bid exchange rules, while experiments 2 and 4 (the “treatment” sessions) were conducted under oral-bid rules (in each case only buyers made quotations), and (2) the subjects were different in the two sessions, but were obtained from the same population (Caltech undergraduates).

Experiments 1 and 3 replicated the findings of Williams in a multiunit posted-bid exchange experiment while experiments 2 and 4 replicated in the multiunit case the results Smith isolated for single-unit oral-bid auctions. We conclude that Williams’ result—contract prices below equilibrium when buyers post bids—is due to the posted-bid institution and has nothing to do with individual traders’ having a capacity to trade a single or several units in each trading period. We conclude that the Smith result—contract prices above equilibrium when buyers make oral bids—extends in its general pattern where subjects have a capacity to trade multiple units per trading period. Our explanation of the robust results of Williams is that the effect of seller discrimination causes buyers to bid lower than in the oral auction. Therefore, the posted-price institution is related analytically to sealed-bid auctions in which discrimination causes a lowering of bids, as demonstrated by the experimental results of Smith [7] and Belovicz [1].

Both the posted-price and the oral-bid markets reveal a high degree of efficiency. The oral bid, however, is consistently near 100 per cent efficiency. Both models of organization have implications about the distribution of income. Income tends to be distributed in *favour* of the side which posts a price bid (offer) and *against* the side which makes an oral price bid (offer).

A single unifying theory which explains all the observed regularities does not yet exist. We attempt to describe the dynamics and offer these results as an empirical challenge for those interested in price adjustment processes.

APPENDIX: INSTRUCTIONS AND PAYMENT SCHEDULES

A.

TABLE III
Subject identification number

Subject number	Demand or supply based on revenue or cost conditions assigned to subject (see figure 2)	Experiment in which subject participated
1	D ₁	1
2	D ₁	1
3	D ₂	1
4	D ₂	1
5	S ₁	1
6	S ₁	1
7	S ₂	1
8	S ₂	1
9	D ₁	2
10	D ₁	2
11	D ₂	2
12	D ₂	2
13	S ₁	2
14	S ₁	2
15	S ₂	2
16	S ₂	2
17	D ₁	3
18	D ₁	3
19	D ₂	3
20	D ₂	3
21	S ₁	3
22	S ₁	3
23	S ₂	3
24	S ₂	3
25	D ₁	4
26	D ₁	4
27	D ₂	4
28	D ₂	4
29	S ₁	4
30	S ₁	4
31	S ₂	4
32	S ₂	4

B. INSTRUCTIONS: EXPERIMENTS 1 AND 2

The instructions used for these two experimental sessions are divided into five parts: General, Specific Instructions to Sellers, Specific Instructions to Buyers, Market Organization, Payoff Tables. The General section was identical for both sessions. The Specific Instructions for Sellers sections and the Specific Instructions for Buyers sections are almost the same³ for both experiments, and the payoff formats were identical. The only major difference occurs in the Market Organization section, and the differences here reflect the changes in the treatment variable.

INSTRUCTIONS

General

This is an experiment in the economics of market decision making. Various research foundations have provided funds for the conduct of this research. The instructions are simple, and if you follow them carefully and make good decisions you might earn a considerable amount of money which will be paid to you in cash after the experiment.

In this experiment we are going to simulate a market in which some of you will be buyers and some of you will be sellers in a sequence of market days or trading periods. Two kinds of sheets will now be distributed—information for buyers and information for sellers. The sheets are identified and numbered. The number is only for data-collecting purposes. If you have received sellers' information, you will function only as a seller in this market. Similarly, if you have received buyers' information, you will function only as a buyer in this market. The information you have received is for your own private use. **Do not reveal it to anyone.**

This is a one commodity market in which there is no product differentiation. That is, each seller produces a product which is similar in all respects to the products offered by the other sellers. A seller is free to sell to any buyer or buyers. Likewise, a buyer may purchase from any seller or sellers.

Specific Instructions for Sellers

During each market period you are free to produce and sell any of the amounts listed on your information sheet. Assume that you produce only for immediate sale—there are no inventories. The dollar amounts listed in column 2 on your information sheet are your costs of producing that quantity.

Your payoffs are computed as follows: At the beginning of the experiment you will receive starting capital of \$0.30. If you are able to make any sales, you will receive the difference between your sales revenue and your cost. For example, if you were to sell two units at \$100 each, total revenue would be \$200. Suppose your information sheet indicated that the cost of producing two units was \$190. Your total profit would then be $\$200 - \$190 = \$10$ for the trading period. If you sold two units for less than \$190 you would incur a loss. Column 3 will be useful to a seller in deciding at any time during a given trading period whether to sell an additional unit. Suppose a seller has already sold one unit at a profit, and wants to know if he should sell a second unit. If the additional cost of producing the second unit is \$10, then he will lose money on that unit if he sells it at any price below \$10. Obviously, these figures are illustrative only and should not be assumed to apply to the actual sellers in this experiment.

All of your profits will be added to your starting capital, and any losses you might incur will be subtracted. Your total payoffs will be accumulated over several trading periods and the total amount will be paid to you after the experiment.

Specific Instructions for Buyers

During each market period you are free to purchase any of the quantities listed on your information sheet. Assume that you are buying this commodity for the purpose of reselling it in an entirely different market. The dollar amounts listed beside each quantity are the total value of that quantity. That is, they are the amounts you can sell that quantity for in the other market.

Your payoffs are computed as follows: You will receive starting capital of \$0.30. If you are able to make any purchases, you will receive the difference between the total value as shown on your information sheet and the total amount you paid for the purchases. For example, if you were to purchase one unit for \$105 and another for \$95, you would obviously have paid a total of \$200 for the two units. Suppose your information sheet indicated that the revenue from two units was \$210. Your profit for the market period would then be $\$210 - \$200 = \$10$. If you had paid more than \$210 for the two units, you would have incurred a loss. Column 3 will be useful to a buyer in deciding at any time during a given period whether to buy an additional unit. Suppose a buyer has already bought two units at a profit, and wants to know if he should buy a third unit. If the additional revenue he gets from the third unit is \$7, then he will lose money on that unit if he buys it at any price above \$7. Obviously, these figures are illustrative only and should not be assumed to apply to the actual buyers in this experiment. All of your profits will be added to your starting capital, and any losses you might incur will be subtracted.

Your total payoffs will be accumulated over several trading periods and the total amount will be paid to you after the experiment.

Market Organization (included in Exp. 1 instructions but not Exp. 2)

The market for this commodity is organized as follows: we open the market for each trading day. Each buyer decides on a purchase price which he will write on one of the cards provided. The buyers will be given two minutes to submit their prices. The cards will be collected and the prices written on the blackboard. Sellers will then be free to make offers to sell whatever quantities they desire and to specify the buyer to whom they wish to sell. Offers will be made as follows: a seller will raise his hand and, when called upon, will state the quantity he wishes to sell and the buyer to whom he wishes to sell. The buyer will then accept any part of the seller's offer by stating the quantity he wishes to buy. However, when a buyer posts a price,

TABLE IV

Incentive schedules: Experiments 1 and 2

Seller Number..... SELLER'S COST SCHEDULE			Seller Number..... SELLER'S COST SCHEDULE		
Quantity of commodity Q in each trading period	Total cost of producing Q units of commodity	Additional cost of producing the last or Q th unit of commodity	Quantity of commodity Q in each trading period	Total cost of producing Q units of commodity	Additional cost of producing the last or Q th unit of commodity
0	\$0.00		0	\$0.00	
1	0.26	\$0.26	1	0.28	\$0.28
2	0.64	0.38	2	0.60	0.32
3	1.14	0.50	3	0.96	0.36
4	1.74	0.60	4	1.38	0.42
5	2.44	0.70	5	1.88	0.50
6	3.22	0.78	6	2.48	0.60
7	4.08	0.86	7	3.18	0.70
8	5.02	0.94	8	4.00	0.82
			9	4.94	0.94
			10	6.00	1.06

Buyer Number..... BUYER'S REVENUE SCHEDULE			Buyer Number..... BUYER'S REVENUE SCHEDULE		
Quantity of commodity Q in each trading period	Total revenue from Q units of commodity	Additional revenue from the last or Q th unit of the commodity	Quantity of commodity Q in each trading period	Total revenue from Q units of commodity	Additional revenue from the last or Q th unit of the commodity
0	\$0.00		0	\$0.00	
1	0.92	\$0.92	1	0.94	\$0.94
2	1.80	0.88	2	1.76	0.82
3	2.64	0.84	3	2.46	0.70
4	3.42	0.78	4	3.06	0.60
5	4.12	0.70	5	3.56	0.50
6	4.72	0.60	6	3.98	0.42
6	5.22	0.50	7	4.32	0.34
8	5.60	0.38	8	4.58	0.26
9	5.86	0.26			
10	6.00	0.14			

he must be prepared to *buy at least one unit* if any seller wishes to sell to him. If a number of sellers desire to make simultaneous offers, one of them will be selected at random and he will then make his desired sales. If the first buyer will not purchase all units the seller wants to sell, the seller is free to make contracts with another buyer or buyers.

When the first seller has made all his contracts, another seller will be selected at random and he will make his desired purchases. The process will be continued until there are no offers to sell. This completes the trading day. We will reopen the market for a new trading day by having buyers submit new prices and the process will be repeated.

Are there any questions?

Market Organization (included in Exp. 2 instructions but not Exp. 1)

The market for this commodity is organized as follows: we open the market for a trading day. Any buyer is free at any time to raise his hand and make a verbal bid to buy one unit of the commodity at a specified price. Any seller is free to accept or not accept the bid of any buyer but sellers cannot make counter offers. If a bid is accepted a binding contract has been closed for a single unit between that buyer and seller.

This process continues for a period of time. You will be warned in advance before the market closes and more bids will be called for before actually closing. This completes a market “day”. We will then reopen the market for a new trading period. The cost and revenue tables apply to each new trading period, and represent cost or revenue per period.

Are there any questions?

C. INSTRUCTIONS: EXPERIMENTS 3 AND 4

Instructions used in Experimental Sessions 3 and 4 are identical with the exception of the section entitled “Market Organization”. The Market Organization sections outline the change in the treatment variable. Both forms are included below.

INSTRUCTIONS

General

This is an experiment in the economics of market decision making. Various research foundations have provided funds for this research. The instructions are simple and if you follow them carefully and make good decisions you might earn a considerable amount of money which will be paid to you in cash.

In this experiment we are going to simulate a market in which some of you will be buyers and some of you will be sellers in a sequence of market days or trading periods. Attached to the instructions you will find a sheet, labelled Buyer or Seller, which describes the value to you of any decisions you might make. **You are not to reveal this information to anyone.** It is your own private information.

Specific Instructions to Buyers

During each market period you are free to purchase from any seller or sellers as many units as you might want. For the first unit that you buy *during a trading period* you will receive the amount listed in row (1) marked *1st unit redemption value*; if you buy a second unit you will receive the additional amount listed in row (5) marked *2nd unit redemption value*; etc. The profits from each purchase (which are yours to keep) are computed by taking the difference between the redemption value and purchase price of the unit bought. *Under no conditions may you buy a unit for a price which exceeds the redemption value.* In addition to this profit you will receive a 5 cent commission for each purchase. That is

$$[\text{your earnings} = (\text{redemption value}) - (\text{purchase price}) + 0.05 \text{ commission}].$$

Suppose for example that you buy two units and that your redemption value for the first unit is \$200 and for the second unit is \$180. If you pay \$150 for your first unit and \$160 for the second unit, your earnings are:

$$\begin{aligned} \$ \text{ earnings from 1st} &= 200 - 150 + 0.05 = 50.05 \\ \$ \text{ earnings from 2nd} &= 180 - 160 + 0.05 = 20.05 \\ \text{total } \$ \text{ earnings} &= 50.05 + 20.05 = 70.10 \end{aligned}$$

The blanks on the table will help you record your profits. The purchase price of the first unit you buy during the first period should be recorded on row (2) *at the time of purchase*. You should then record the profits on this purchase as directed on rows (3) and (4). At the end of the period record the total of profits and commissions on the last row (41) on the page. Subsequent periods should be recorded similarly.

Specific Instructions to Sellers

During each market period you are free to sell to any buyer or buyers as many units as you might want. The first unit that you sell *during a trading period* you obtain at a cost of the amount listed on the attached sheet in the row (2) marked *cost of 1st unit*; if you sell a second unit you incur the cost listed in the row (6) marked *cost of the 2nd unit*; etc. The profits from each sale (which are yours to keep) are computed by

Unit Sold	Trading Period Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1st unit redemption value	.92	.92	.92	.92	.92	.92	.92	.92	.92	.92	.92	.92	.92	.92	.92
	Purchase price															
	Profit (row 1 - row 2)															
	Profit + 5 cents commission															
2	2nd unit redemption value	.88	.88	.88	.88	.88	.88	.88	.88	.88	.88	.88	.88	.88	.88	.88
	Purchase price															
	Profit (row 5 - row 6)															
	Profit + 5 cents commission															
3	3rd unit redemption value	.84	.84	.84	.84	.84	.84	.84	.84	.84	.84	.84	.84	.84	.84	.84
	Purchase price															
	Profit (row 9 - row 10)															
	Profit + 5 cents commission															
4	4th unit redemption value	.78	.78	.78	.78	.78	.78	.78	.78	.78	.78	.78	.78	.78	.78	.78
	Purchase price															
	Profit (row 13 - row 14)															
	Profit + 5 cents commission															
5	5th unit redemption value	.70	.70	.70	.70	.70	.70	.70	.70	.70	.70	.70	.70	.70	.70	.70
	Purchase price															
	Profit (row 17 - row 18)															
	Profit + 5 cents commission															
6	6th unit redemption value	.60	.60	.60	.60	.60	.60	.60	.60	.60	.60	.60	.60	.60	.60	.60
	Purchase price															
	Profit (row 21 - row 22)															
	Profit + 5 cents commission															
7	7th unit redemption value	.50	.50	.50	.50	.50	.50	.50	.50	.50	.50	.50	.50	.50	.50	.50
	Purchase price															
	Profit (row 25 - row 26)															
	Profit + 5 cents commission															
8	8th unit redemption value	.38	.38	.38	.38	.38	.38	.38	.38	.38	.38	.38	.38	.38	.38	.38
	Purchase price															
	Profit (row 29 - row 30)															
	Profit + 5 cents commission															
9	9th unit redemption value	.26	.26	.26	.26	.26	.26	.26	.26	.26	.26	.26	.26	.26	.26	.26
	Purchase price															
	Profit (row 33 - row 34)															
	Profit + 5 cents commission															
10	10th unit redemption value	.14	.14	.14	.14	.14	.14	.14	.14	.14	.14	.14	.14	.14	.14	.14
	Purchase price															
	Profit (row 37 - row 38)															
	Profit + 5 cents commission															
	Total per period earnings															
	Damages															
	Net income															
	Capital payment															

Name _____ Soc. Sec. No. _____ Total Payment _____

Address _____

FIGURE 10
Record of Purchases and Earnings, Buyers No. 1 & 2

taking the difference between the price at which you sold the unit and the cost of the unit. *Under no conditions may you sell a unit at a price below the cost of the unit.* In addition to this profit you will receive a 5 cent commission for each sale. That is

$$[\text{your earnings} = (\text{sale price of unit}) - (\text{cost of unit}) + (0.05 \text{ commission}).]$$

Your total profits and commissions for a trading period, which are yours to keep, are computed by adding up the profit and commissions on sales made during the trading period.

Suppose, for example, your cost of the 1st unit is \$140 and your cost of the second unit is \$160. For illustrative purposes we will consider only a two-unit case. If you sell the first unit at \$200 and the second unit at \$190, your earnings are:

$$\begin{aligned} \$ \text{ earnings from 1st} &= 200 - 140 + 0.05 = 60.05 \\ \$ \text{ earnings from 2nd} &= 190 - 160 + 0.05 = 30.05 \\ \text{total } \$ \text{ earnings} &= 60.05 + 30.05 = 90.10 \end{aligned}$$

The blanks on the table will help you record your profits. The sale price of the first unit you sell during the 1st period should be recorded on row (1) *at the time of sale*. You should then record the profits on this sale as directed on rows (3) and (4). At the end of the period record the total of profits and commissions on the last row (41) on the page. Subsequent periods should be recorded similarly.

Market Organization (included in instructions for Exp. 3 but not Exp. 4)

The market for this commodity is organized as follows: we open the market for each trading day. Each buyer decides on a purchase price which he will write on one of the cards provided. The buyers will be given two minutes to submit their prices. The cards will be collected and the prices written on the blackboard. Sellers will then be free to make offers to sell whatever quantities they desire and to specify the buyer to whom they wish to sell. Offers will be made as follows: a seller will be chosen using random numbers, and will state the quantity he wishes to sell and the buyer to whom he wishes to sell. The buyer will then accept any part of the seller's offer by stating the quantity he wishes to buy. However, when a buyer posts a price, he must be prepared to *buy at least one unit*. If the first buyer will not purchase all units the seller wants to sell, the seller is free to choose a second buyer, and so on.

When the first seller has made all his contracts, another seller will be selected at random and he will make his desired purchases. The process will be continued until there are no offers to sell. This completes the trading day. We will reopen the market for a new trading day by having buyers submit new prices and the process will be repeated. Except for the offers and their acceptance you are not to speak to any other subject. You are free to make as much profit as you can.

Are there any questions?

Market Organization (included in instructions for Exp. 4 but not Exp. 3)

The market for this commodity is organized as follows: we open the market for a trading period (a trading "day"). The period lasts for minutes. Any buyer is free at any time during the period, to raise his hand and make a verbal bid to buy one unit of the commodity at a specified price. Any seller is free to accept or not accept the bid of any buyer but *sellers cannot make counter offers*. If a bid is accepted a binding contract has been closed for a single unit and the buyer and seller will record the contract price to be included in their earnings. Any ties in bids or acceptances will be resolved by a random choice of buyer or seller. Except for the bids and their acceptance you are not to speak to any other subject. There are likely to be many bids that are not accepted, but you are free to keep trying, and as a buyer or a seller you are free to make as much profit as you can.

Are there any questions?

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NOTES

1. The behaviour on the supply side could be a point where the analogy fails. In the case of a discriminative sealed-bid auction the bidder states the maximum quantity he will accept along with his price bid. Then the supplier, following the institutional convention, sells to the highest bidder the amount the bidder desires and moves on to the next highest bidder, etc. This continues until the total quantity offered is exhausted. There is no attempt on the part of the supplier to act strategically by failing to supply all that is requested at a price (and thus influence prices) or by favouring in quantity those buyers who have historically offered higher prices. The reader may recognize this lack of strategic behaviour as the competitive supply response. Indeed, if we can depend upon competitive responses by suppliers in the posted-bid auction, the analogy between the two institutions would appear to be close.

2. Data regarding all bids, offers and individual contracts are available from the authors.

3. The instructions are those used in experimental session 2. Those used in experiment 1 are identical to these after the following sentences are deleted. In the Specific Instructions to Sellers section delete: "Column 3 will be useful to a seller deciding at any time during a given trading period whether to sell an additional unit. Suppose a seller has already sold one unit at a profit and wants to know if he should sell a second unit. If the additional cost of producing the second unit is \$10, then he will lose money on that unit if he sells it at any price below \$10." In the Specific Instructions to Buyers section delete: "Column 3 will be useful to a buyer in deciding at any time during a given period whether to buy an additional unit. Suppose a buyer has already bought two units at a profit, and wants to know if he should buy a third unit. If the additional revenue he gets from the third unit is \$7, then he will lose money on that unit if he buys it at any price above \$7."

REFERENCES

- [1] Belovicz, Mayer W. "The Sealed-Bid Auction: Experimental Studies" (PhD dissertation, Purdue University, August 1967).
- [2] Hess, Alan C. "Experimental Evidence on Price Formation in Competitive Markets", *Journal of Political Economy*, **80**, 2 (March/April 1972), 375-385.
- [3] Hurwicz, Leonid. "The Design of Mechanism for Resource Allocation", *American Economic Review*, **63**, 2 (May 1973), 1-30.
- [4] Rothschild, Michael. "Models of Market Organization with Imperfect Information: A Survey", *Journal of Political Economy*, **81**, 6 (November/December 1973), 1283-1308.
- [5] Smith, Vernon L. "An Experimental Study of Competitive Market Behavior", *Journal of Political Economy*, **70**, 2 (April 1962), 111-137.
- [6] Smith, Vernon L. "Effect of Market Organization on Competitive Equilibrium", *Quarterly Journal of Economics*, **78**, 94 (May 1964), 181-201.
- [7] Smith, Vernon L. "Experimental Studies of Discrimination Versus Competition in Sealed-Bid Auction Markets", *Journal of Business*, **40**, 1 (January 1967), 56-84.
- [8] Smith, Vernon L. "Experimental Economics: Induced Value Theory", *American Economic Review*, **66**, 2 (May 1976).
- [9] Williams, Fred. "Effect of Market Organization on Competitive Equilibrium: The Multiunit Case", *Review of Economic Studies*, **40** (January 1973), 97-113.